DBMS TASK 4

Choosing Between SQL and NoSQL Data Stores

Purpose

This comparison outlines the strengths of relational and NoSQL systems and explains how application needs influence database selection.

A. Technology Comparison

Dimension	Relational / SQL	NoSQL
Organisation	Structured tables linked by keys.	Documents, key-value records, graphs, or wide-column data.
Schema approach	Columns and types are defined in advance.	Often flexible, depending on the selected product.
Scaling	Vertical scaling is common; distributed options also exist.	Many products are built to distribute data across multiple nodes.
Consistency	Transactions and strong consistency are common features.	Guarantees vary; some products provide strong consistency and transactions.
Queries	SQL supports joins, grouping, and complex transactional queries.	Query patterns depend on the NoSQL model and product.
Best fit	Structured records, reliable transactions, and connected entities.	Flexible records, high-volume distributed workloads, or specialised data models.
Examples	MySQL, PostgreSQL, Microsoft SQL Server.	MongoDB, Cassandra, Redis.

B. Examples in Practice

Relational example: online orders

An online retailer can store customer profiles, product listings, order headers, order lines, and payment details in MySQL. Foreign keys link the records, while transactions help keep related changes consistent.

NoSQL example: flexible product catalogue

A catalogue may contain items with different sets of attributes. A document database such as MongoDB can store laptop specifications in one document and clothing size or fabric details in another without requiring every product to share identical columns.

C. What Drives the Decision?

Teams consider the data shape, transaction guarantees, expected read and write volume, latency targets, scaling plans, maintenance skills, cost, and integration requirements. A database should be selected for the workload rather than because one technology is universally superior.

D. Selection for OnlineShoppingDB

MySQL is appropriate for this project because the core records are structured and linked: orders belong to customers, order lines reference products, and payments reference orders. Relational constraints make these links easier to validate.

Closing Remarks

SQL and NoSQL technologies offer different trade-offs. A good selection matches the data model, consistency needs, query patterns, and growth expectations of the application.

SQL EXECUTION NOTE

Compare relational and NoSQL databases in the written report.

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-- Assignment 4 is a theoretical comparison.
-- No SQL commands are required for this assignment.
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